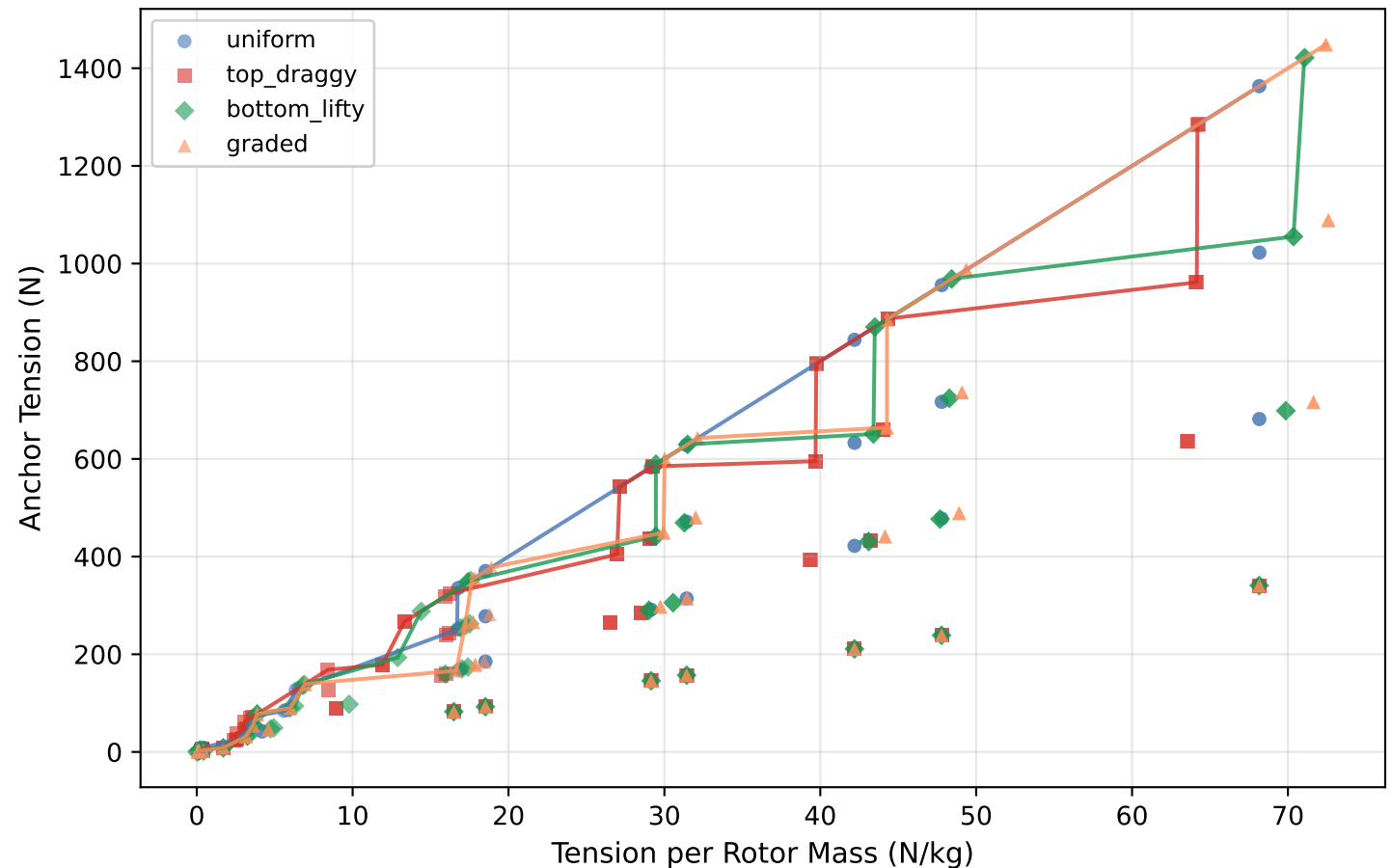


## Pareto: Tension vs Mass Efficiency by Tilt Profile BEM v2.1 — All (R,N,elevation) Combinations



*IMPLICATIONS: The tilt profile spread is small — all four profiles occupy overlapping regions of the Pareto space. Graded and top-draggy profiles reach the highest absolute tensions (>800 N) while uniform and bottom-lifty cluster at lower tensions. The Pareto frontier for each profile shows the same trade-off: higher N/kg means lower absolute tension. No single profile dominates — engineering choice depends on whether you prioritise mass efficiency (bottom-lifty) or peak tension (graded/top-draggy). Data: BEM v2.1, 384 configs.*